

# Unit 2 Pre-Test

## Scientific Notation

Standards assessed: NC.8.EE.3, NC.8.EE.4

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

### SCORE **3.0** Basic Understanding

**NC.8.EE.3** Express very large and very small quantities in scientific notation and standard form.

- I can write numbers in scientific notation and in standard form, and recognize correct form.

Match each vocabulary term to its definition, then write each number as directed.

|                                                    |                                                                                |
|----------------------------------------------------|--------------------------------------------------------------------------------|
| <input type="text" value="—"/> scientific notation | <b>A.</b> A number written the ordinary way, such as 45,000.                   |
| <input type="text" value="—"/> coefficient         | <b>B.</b> The decimal factor in $a \times 10^n$ — at least 1 and less than 10. |
| <input type="text" value="—"/> standard form       | <b>C.</b> Shows a small number between 0 and 1 (move the decimal left).        |
| <input type="text" value="—"/> base                | <b>D.</b> A number written as $a \times 10^n$ , where $1 \leq a < 10$ .        |
| <input type="text" value="—"/> positive exponent   | <b>E.</b> In a power of ten, the 10 that is multiplied repeatedly.             |
| <input type="text" value="—"/> negative exponent   | <b>F.</b> Shows a large number, 10 or greater (move the decimal right).        |

Parts of a number in scientific notation — reference

$$3.4 \times 10^5$$

|                                    |             |                                   |
|------------------------------------|-------------|-----------------------------------|
| <b>coefficient</b>                 | <b>base</b> | <b>exponent</b>                   |
| the decimal factor (1 to under 10) | always 10   | how many places the decimal moves |

A positive exponent makes a **large** number; a negative exponent makes a **small** one.

1. Write in scientific notation

45,000

Answer:

2. Write in standard form

$3 \times 10^4$

Answer:

3. Write in scientific notation

0.0062

Answer:

4. Write in standard form

$7.1 \times 10^{-2}$

Answer:

5. Write in scientific notation

580,000

Answer:

6. Fix the notation

Is  $12.5 \times 10^6$  correct? Write the correct form.

Answer:

## SCORE 4.0 Target Content

**NC.8.EE.4** Perform operations with numbers in scientific notation and compare their sizes.

• I can multiply and divide numbers in scientific notation and tell how many times larger one is than another.

**Multiply, divide, or compare. Write each answer in correct scientific notation.**

7. Multiply

$(3 \times 10^4)(2 \times 10^3)$

Answer:

8. Divide

$(8 \times 10^9) \div (4 \times 10^5)$

Answer:

9. Multiply

$(5 \times 10^6)(4 \times 10^3)$

10. Compare

How many times larger is  $6 \times 10^8$  than  $2 \times 10^3$ ?

Answer: \_\_\_\_\_

Answer: \_\_\_\_\_

11. Divide

$$(9 \times 10^{-2}) \div (3 \times 10^4)$$

Answer: \_\_\_\_\_

12. Multiply

$$(2.5 \times 10^5)(2 \times 10^{-8})$$

Answer: \_\_\_\_\_

## SCORE 5.0 Advanced Content

**NC.8.EE.4** Add and subtract in scientific notation and solve multi-step real-world problems.

• I can add and subtract in scientific notation and solve real-world problems, writing answers in scientific notation.

**Solve. Write each answer in correct scientific notation.**

13. Add

$$(4 \times 10^5) + (3 \times 10^5)$$

Answer: \_\_\_\_\_

14. Add

$$(5 \times 10^6) + (3 \times 10^5)$$

Answer: \_\_\_\_\_

15. Subtract

$$(8.4 \times 10^{-3}) - (2.1 \times 10^{-3})$$

Answer: \_\_\_\_\_

16. Apply

A proton's mass  $\approx 1.67 \times 10^{-27}$  kg.

Find the mass of 2,000 protons.

Answer: \_\_\_\_\_

17. Apply

Light travels  $3 \times 10^8$  m/s. How far in  $2 \times 10^2$  seconds?

Answer: \_\_\_\_\_

18. Apply

A server holds  $4.5 \times 10^{12}$  bytes. Each video is  $9 \times 10^5$  bytes. How many videos fit?

Answer: \_\_\_\_\_

